

HALO COLLAR × EDGE IMPULSE

Collaboration Plan

Pet Activity Detection | Executive Overview

The Challenge

97.6%

Lab Data Accuracy

Validated



~60%

Real-World Data Accuracy

37-point gap

Preliminary Root Causes

Collar orientation variability	Features break when collar rotates from assumed position
Firmware vs. offline pipeline gaps	Real-time data accumulation may differ from collection mode
Controlled-environment training bias	Lab data does not reflect natural behavior diversity
Class taxonomy complexity	8 classes including hard-to-distinguish and minority categories

Engagement Structure & Investment

12-month platform + engineering support agreement

\$133,000

Platform + Advanced Premium Support

12 months

\$8K–\$30K

Commercial Deployment License

Scaled to device volume

Payment Schedule

Phase	Payment	Kick off	Projected activities (including but not limited to)
Phase 1	\$66,500	At execution	• System analysis, pipeline audit & foundational review • Data strategy, class simplification & collection protocols • Initial model development, architecture exploration & evaluation
Phase 2	\$66,500	Week 16	• Model optimization, accuracy fine-tuning & advanced techniques • On-device deployment, parity validation & firmware integration • Field testing, knowledge transfer & ongoing support

Success Criteria

Targets to be validated and refined after Phase 1 discovery. Not contractual guarantees.

Metric	Current	Phase 1 Target	Phase 2 Target
Real-world data accuracy	~60%	80–85%	90%+
Collar orientation robustness	■	■	■
Cloud-to-device parity	■	■	■
Model footprint	51KB / 1.1KB	Within the budget	Within the budget

Target Outcome

Improvement of real-world data test accuracy within 16 weeks, with clear path for improvement toward 90%+ during Phase 2 activities. Based on experience with comparable pet activity detection engagements.

Our Approach

Each cycle of experimentation reveals new insights that reshape our approach. We do not follow a linear checklist. Instead, findings from modeling feed back into data strategy, pipeline discoveries reshape the analysis, and evaluation results drive the next round of experiments.

Analyze → Experiment → Evaluate → Refine → Repeat

Phase 1: Understand, Then Restructure

↻ Iterative by design: these pillars are not sequential steps. Findings from each feed back into the others through multiple experiment-evaluate-refine cycles.



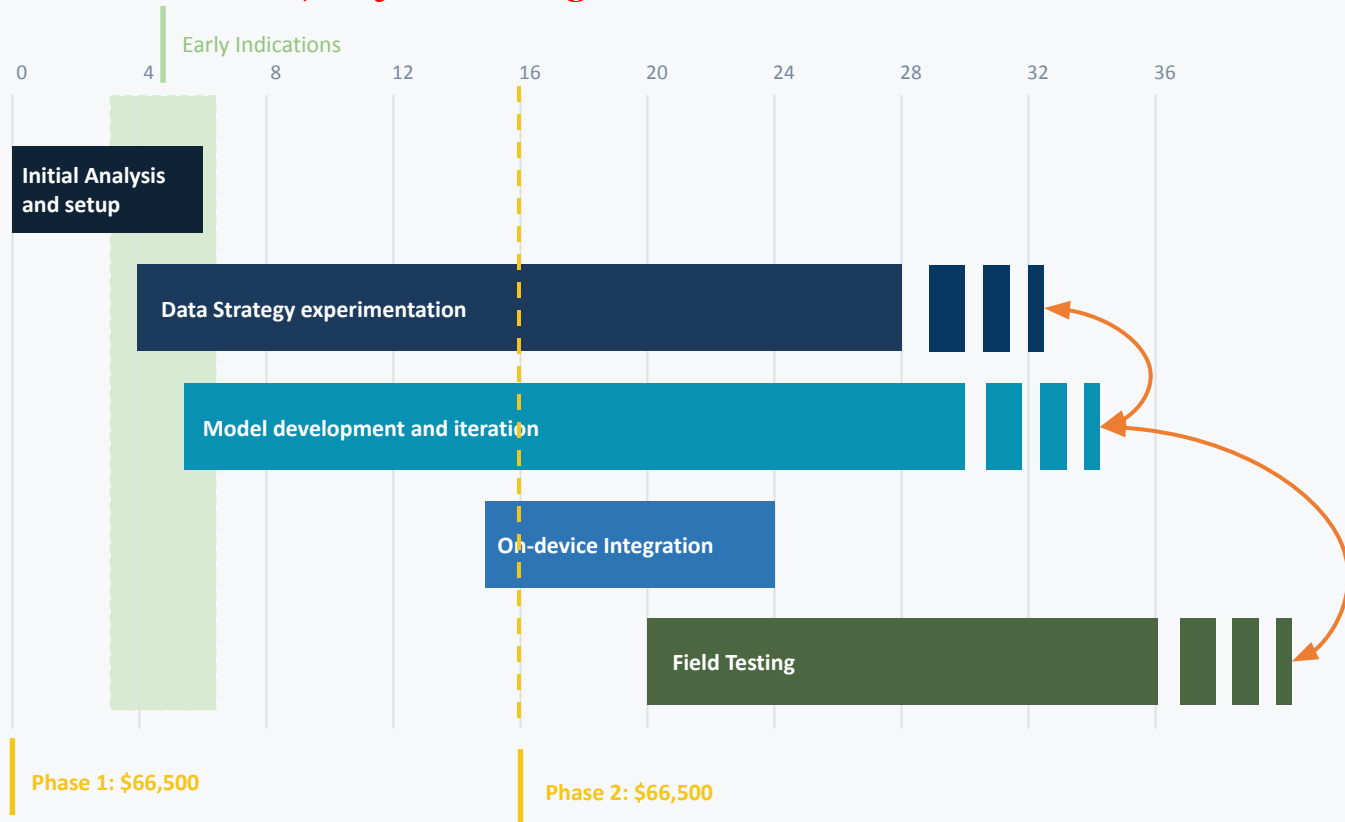
Phase 2: Optimize, Validate & Deploy

🔄 Optimization is iterative: each accuracy gain informs the next round of data collection, model tuning, and field validation.

Data Strategy	Model Optimization	On-Device Integration	Field Testing
- Expanded real-world data collection targeting failure modes - Breed & size diversity campaigns - Refined augmentation based on Phase 1 results - Iterative label quality improvement	- Fine-tune best-performing architecture from Phase 1 - Hyperparameter & training optimization - Advanced techniques: ensemble, TCN, multi-task learning - Per-class accuracy refinement for weak classes - Temporal smoothing & confidence gating tuning	- Deploy optimized pipeline to STM32 collar - Cloud-to-device parity validation - Firmware adjustments as needed - Integration documentation - Knowledge transfer sessions	- Structured real-world testing with labeled ground truth - Performance benchmarking across breed categories - Edge-case and failure mode identification - Results feed back into model optimization cycle
Data drives accuracy gains	Push toward 90%+ accuracy	Production-ready deployment	Real-world validation loop

High Level Timeline

Notional timeline, subject to change



Risks & How We've Prepared

Risk	Why It Matters	How We've Prepared
Data collection timing	Diverse real-world data drives improvement.	Halo data lead by Week 2. Regular check-ins.
Data quality	Labeling issues may exist in current dataset.	Data audit. Edge Impulse analysis tools.
Orientation	Software may not fully solve collar rotation.	Quantify impact. Alternate pre-processing approach / attempt models that are invariant.
Device constraints	New architectures must fit STM32 limits.	Hardware-aware pipeline from day one for each step of the process.

What We Need from Halo

Requirement	Details	When
Engineering Access	Named ML/embedded lead with decision authority	Week 1 ->
Data Access	Full training data, QA results, field recordings	Week 1 ->
Hardware	A device prototype(s) + firmware for on-device testing	Week 1 ->
Data Collection	Execute collection campaigns per agreed protocols, real world data on demand	Throughout, at scale by start of Phase 2
QA/Field Testing	Structured field tests with labeled ground truth	Throughout

Collaboration Cadence

Touchpoint	Frequency	Purpose / Format
Engineering Sync	1x/week	Hands-on technical collaboration, experiment review, blocker resolution.
Stakeholder Review	Bi-weekly	Progress report to Halo leadership.
Quarterly Business Review	Quarterly	Executive-level review of progress, ROI analysis, strategic direction, and next-quarter planning.

Next Steps

We are ready to begin as soon as the agreement is signed.

- 1 Named Technical Lead assigned and introduced to Halo team
- 2 Kickoff meetings with ML and embedded engineering teams
- 3 Data ingestion into Edge Impulse platform begins
- 4 Foundational pipeline review commences (Phase 1)

The 37-point gap between lab and real-world is solvable. Let's get started.

High Level Timeline

Notional timeline, subject to change

